(ii) allow $10^{-3} \rightarrow 10^{-6}$ m **B1 [2]**

3 (a) constant gradient/straight line **B1 [1]**

(b) (i) 1.2 s **A1**

(ii) 4.4 s **A1 [2]**

(c) *either* use of area under line *or* *h* = average speed x time **C1**

h = ½ x (4.4 – 1.2) x 32 **C1**

= 51.2 m **A1 [3]**

(allow 2/3 marks for determination of *h* = 44 m or *h* =58.4 m
allow 1/3 marks for answer 7.2 m)

(d) $\Delta p = m\Delta v$ OR $p = mv$ **C1**

$= 0.25 \times (28 + 12)$ **C1**

= 10 N s **A1 [3]**

(answer 4 N s scores 2/3 marks)

3 (e) (i) total/sum momentum before = total/sum momentum after **B1**

in any closed system **B1 [2]**

(ii) *either* the system is the ball and Earth **B1**

momentum of Earth changes by same amount **B1**

but in the opposite direction **B1**

or Ball is not an isolated system/there is a force on the ball (B1)

<u>Gravitational</u> force acts on the ball (B1)

causes change in momentum/law does not apply here (B1) **[3]**

(if explains in terms of air resistance, allow first mark only)

4 (a) [illegible] less than 1.50 **B1 [1]**